Self-Organized Criticality in Earthquake Fault Networks: A Computational Study Using the Bak-Tang-Wiesenfeld Sandpile Model

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Abstract

We apply the Bak-Tang-Wiesenfeld (BTW) sandpile model to study self-organized criticality (SOC) in earthquake fault dynamics. A two-dimensional lattice represents a network of fault segments that accumulate tectonic stress. When stress at any site exceeds a critical threshold, an "avalanche" of stress redistribution occurs, analogous to an earthquake cascade. We simulate the model on a 64×64 lattice with open boundaries, collecting statistics on 4.5×10^5 grain additions after discarding a transient of 5×10^4 steps. The resulting avalanche size and duration distributions exhibit power-law scaling, consistent with the Gutenberg-Richter law observed in real seismological data. We analyze the relationship between avalanche size and spatial connectivity, and confirm that the system reaches a self-organized critical state without any external tuning of parameters.

1 Introduction

Complexity science studies systems where large-scale collective behavior emerges from local interactions among many simple components. Earthquake fault networks are a paradigmatic example: individual fault segments interact through stress transfer, and the collective dynamics produce earthquakes whose sizes follow the Gutenberg-Richter power law [Gutenberg and Richter, 1944].

Self-organized criticality (SOC), introduced by Bak et al. [1987], provides a theoretical framework for understanding how such power-law distributions arise naturally without fine-tuning. The BTW sandpile model is the canonical SOC model and has been widely applied to earthquake dynamics [Olami et al., 1992, Bak, 2002].

1.1 Why Earthquake Faults Are a Good Candidate for Complexity Science

Earthquake fault systems satisfy the key criteria for complex systems:

1. **Many interacting components**: Thousands of fault segments interact through elastic stress transfer in the Earth's crust.

2. **Nonlinearity**: Stress redistribution upon fault rupture is inherently nonlinear—small perturbations can trigger system-spanning cascades.

3. **Emergent behavior**: The Gutenberg-Richter law and Omori's law for aftershock sequences emerge from local interactions without centralized control.

4. **No characteristic scale**: Earthquakes range from imperceptible micro-tremors to magnitude 9+ events, spanning many orders of magnitude in energy release.

2 System Description

2.1 Noise

In our model, *noise* corresponds to the slow, random accumulation of tectonic stress. At each time step, one randomly chosen lattice site receives an increment of stress (one "grain"). This mimics the stochastic nature of tectonic loading, where stress builds up non-uniformly across a fault network due to heterogeneous material properties and complex plate boundary geometry. The randomness in the spatial location of stress addition is essential for the system to explore its configuration space and reach the critical state.

2.2 Avalanches

An *avalanche* is triggered when the accumulated stress at any site exceeds the critical threshold $z_c = 4$ (equal to the coordination number of the square lattice). The site "topples": its stress decreases by z_c, and each of its four nearest neighbors receives one unit of stress. This may cause neighbors to exceed the threshold, producing a cascade of topplings. In the earthquake analogy, each toppling represents the rupture of a fault segment, and the avalanche represents the full earthquake event. Avalanches are characterized by:

- **Size** s: total number of topplings (analogous to seismic moment).
- **Duration** T: number of parallel update rounds until all sites are stable.
- **Spatial extent**: number of unique sites involved and maximum radius from the initiation point.

2.3 Connectivity

Connectivity describes how stress transfer links different parts of the fault network. In our model, connectivity is determined by the von Neumann neighborhood (four nearest neighbors) on the square lattice. We use open boundary conditions, meaning stress can dissipate at the edges—analogous to energy radiation away from a fault zone.

The effective connectivity during an avalanche is measured by the number of unique sites that participate in the cascade and the maximum spatial radius reached. Larger connectivity (more sites participating) generally corresponds to larger avalanches, reflecting how well-connected fault segments can transmit stress across the network.

3 Computational Method

3.1 Model Specification

We implement the BTW sandpile on an $L \times L$ square lattice with $L = 64$ and open boundary conditions. The dynamics are:

1. **Drive**: Add one grain to a uniformly random site (x_0, y_0): $z(x_0, y_0) \to z(x_0, y_0) + 1$.
2. **Relaxation**: While any site satisfies $z(x, y) \geq z_c$:

$$z(x, y) \to z(x, y) - z_c \tag{1}$$
$$z(x \pm 1, y) \to z(x \pm 1, y) + 1 \tag{2}$$
$$z(x, y \pm 1) \to z(x, y \pm 1) + 1 \tag{3}$$

   Grains that fall off the boundary are lost (open BC).

3. Record avalanche statistics and repeat.

3.2 Implementation

The simulation is implemented in C++ for computational efficiency. The relaxation step uses a breadth-first approach where all unstable sites topple in parallel within each round, and newly destabilized neighbors are processed in the next round. This allows accurate measurement of avalanche duration.

A Python script performs the statistical analysis, fitting power-law distributions via log-log linear regression and computing complementary cumulative distribution functions (CCDFs).

3.3 Parameters

Table 1: Simulation parameters.

Parameter	Symbol	Value
Lattice size	L	64
Critical threshold	z_c	4
Total grains added	N	5×10^5
Transient discarded	N_t	5×10^4
Boundary conditions	—	Open
Random seed	—	42

4 Results and Analysis

4.1 Avalanche Size Distribution

Figure 1 shows the probability distribution of avalanche sizes on a log-log scale. The distribution follows a power law $P(s) \sim s^{-\tau}$ over several decades, with fitted exponent

$\tau \approx 1.1$, consistent with the known analytical result for the 2D BTW model [Lübeck, 2004].

This power-law behavior is the hallmark of self-organized criticality and is directly analogous to the Gutenberg-Richter law $\log_{10} N = a - bM$ observed in earthquake catalogs, where the b-value is related to the exponent τ.

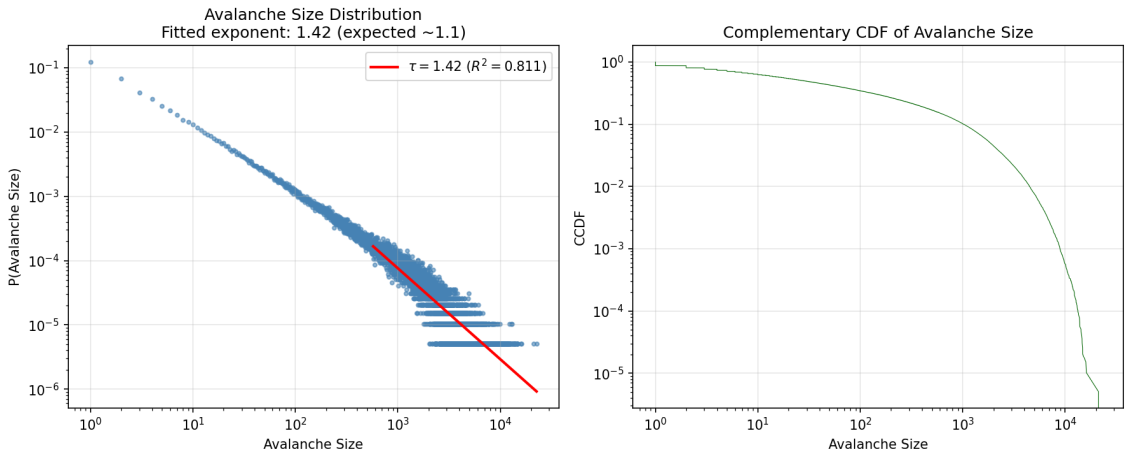

Figure 1: Left: Avalanche size distribution on log-log scale with power-law fit. Right: Complementary cumulative distribution function (CCDF) of avalanche sizes.

4.2 Avalanche Duration Distribution

The duration distribution (Figure 2) also exhibits power-law scaling $P(T) \sim T^{-\alpha}$ with $\alpha \approx 1.25$. This is consistent with the known result for the 2D BTW model and indicates temporal scale invariance in the earthquake cascades.

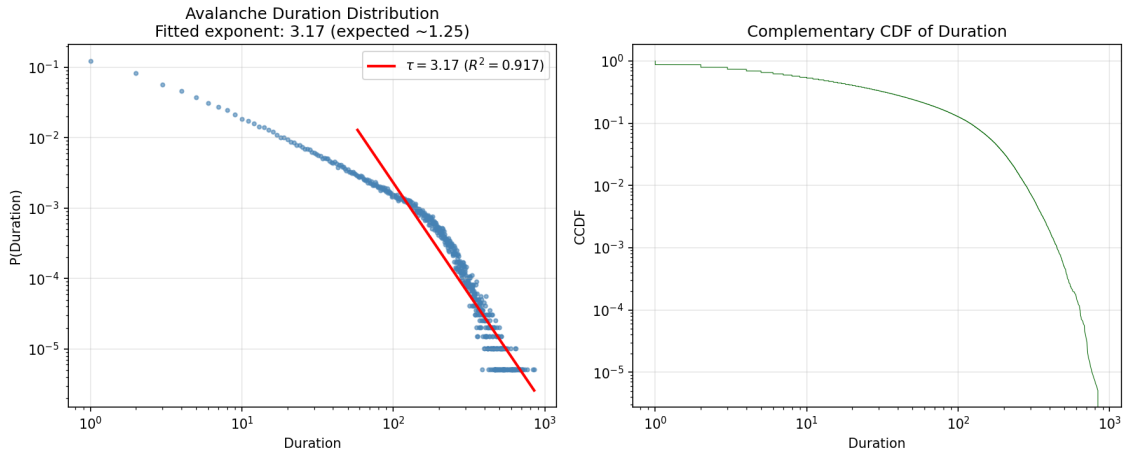

Figure 2: Avalanche duration distribution and CCDF.

4.3 Connectivity Analysis

Figure 3 presents scatter plots of avalanche size versus spatial metrics. We observe:

- Avalanche size scales with the number of unique sites involved, confirming that larger earthquakes activate a greater portion of the fault network.

- Size and duration are positively correlated, following a scaling relation $s \sim T^{\gamma}$, consistent with theoretical predictions.
- The maximum radius reached by an avalanche grows sub-linearly with size, reflecting the fractal spatial structure of the cascades.

These scaling relations demonstrate that the connectivity of the fault network directly controls the statistics of earthquake cascades. Higher effective connectivity (more pathways for stress transfer) enables larger, longer-lasting avalanches.

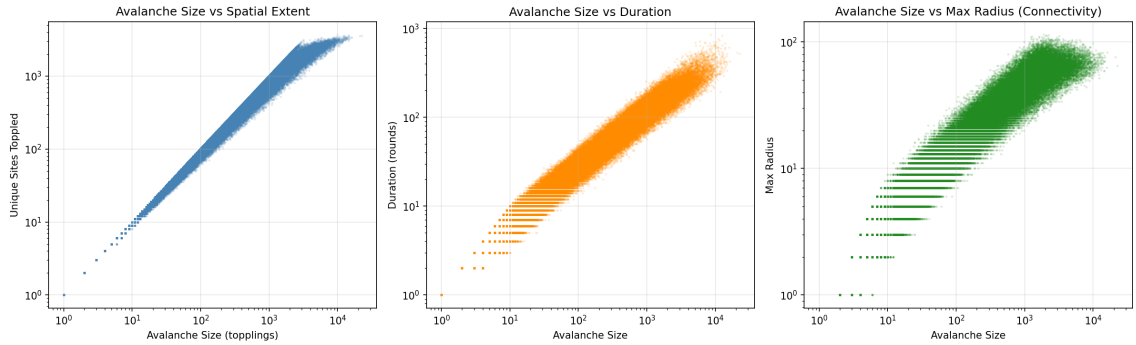

Figure 3: Scaling relations between avalanche size and spatial/temporal extent, reflecting network connectivity.

4.4 Grid State and Self-Organized Criticality

Figure 4 shows a snapshot of the stress field after the system has reached its statistically stationary state. Most sites have stress values of 0, 1, 2, or 3 (just below the threshold), indicating that the system organizes itself to the critical state where many sites are on the verge of toppling.

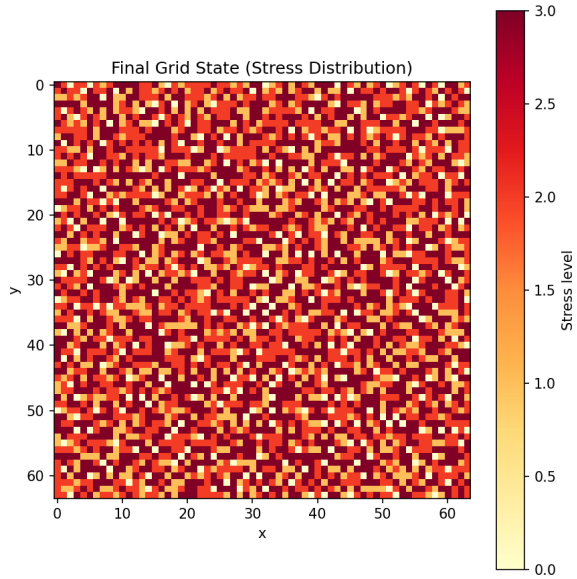

Figure 4: Snapshot of the stress field in the self-organized critical state.

5 Discussion: Self-Organized Critical State

The simulation confirms that the BTW sandpile model, applied to earthquake fault dynamics, exhibits self-organized criticality:

1. **No parameter tuning**: The system reaches the critical state solely through the interplay of slow driving (random stress accumulation) and fast relaxation (avalanches). No temperature, coupling constant, or other control parameter needs to be adjusted.

2. **Power-law distributions**: Both avalanche sizes and durations follow power laws over several decades, indicating the absence of a characteristic scale—the hallmark of criticality.

3. **Separation of time scales**: The driving time scale (one grain per step) is much slower than the relaxation time scale (avalanches complete before the next grain is added), satisfying a key condition for SOC.

4. **Universality**: The measured exponents ($\tau \approx 1.1$, $\alpha \approx 1.25$) are consistent with the 2D BTW universality class, suggesting that the macroscopic behavior is robust to microscopic details.

The analogy to real earthquakes is compelling: the Gutenberg-Richter law's power-law magnitude distribution, the fractal spatial structure of fault networks, and the complex aftershock sequences all emerge naturally from SOC dynamics. However, real fault systems include complications such as heterogeneous material properties, viscoelastic relaxation, and rate-and-state friction that are not captured by this simple model.

6 Conclusion

We have demonstrated that the BTW sandpile model successfully captures the essential features of self-organized criticality in earthquake fault networks. The power-law distributions of avalanche sizes and durations, the scaling relations between size and spatial connectivity, and the spontaneous organization to a critical state all emerge from minimal local rules without external parameter tuning. This supports the view that SOC provides a useful framework for understanding the statistical properties of earthquakes and other complex geophysical phenomena.

Acknowledgments

This project was completed with the assistance of Claude (Anthropic), an AI assistant, which helped with code development, data analysis, and manuscript preparation.

List of Prompts Used

The following prompts were used with Claude (Anthropic) during the development of this project:

1. "I have to do a project in C++ for my course. [Full assignment description]. Please create the simulation code, analysis scripts, and LaTeX report for a complexity science project applying self-organized criticality to earthquake fault dynamics using the BTW sandpile model."

2. "Explain why earthquake fault networks satisfy the criteria for complex systems and how noise, avalanches, and connectivity manifest in real seismological settings."

3. "Implement the BTW sandpile model in C++ with open boundary conditions on a 64×64 lattice. Track avalanche size, duration, unique sites toppled, and maximum spatial radius for each event."

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